

Anticipatory Hand Glove: Understanding Human Actions for Enhanced Interaction

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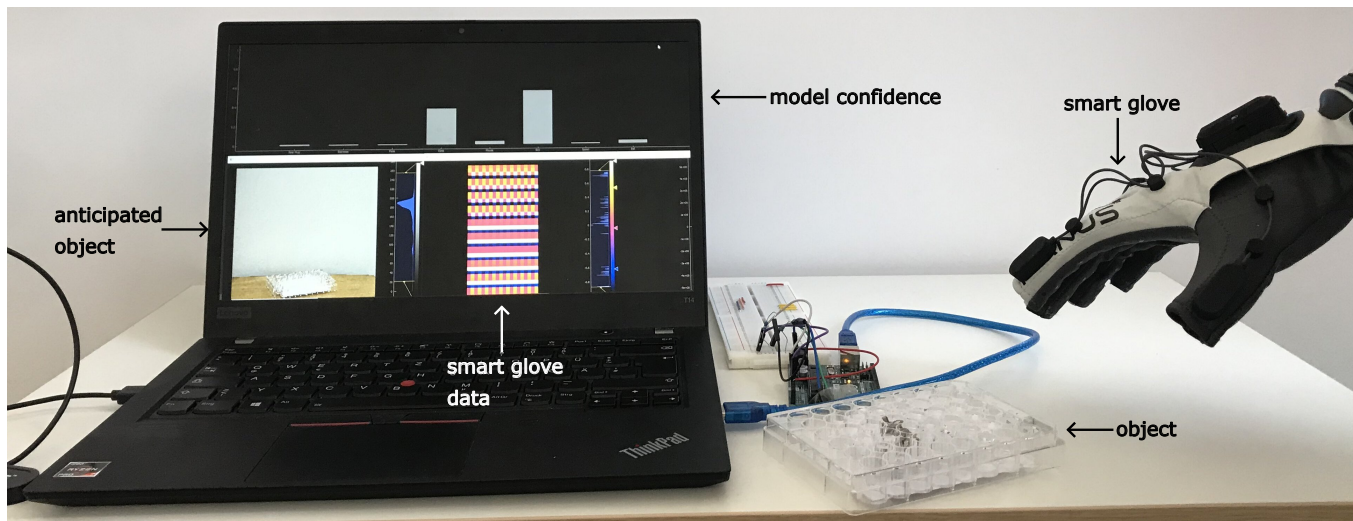


Figure 1: Demo and Experimental Setup for anticipating human intents of grasping various objects from hand gestures by using smart gloves.

ABSTRACT

Perceived real-time interactions between humans and a metaverse often require synchronizing actions in the virtual and real world. Latency is a main blocking factor due to processing and transmitting data over long distances because cloud data centres that deploy

metaverse locate far away. We present an anticipatory computing solution to predict precisely human actions grasping objects based on hand kinesthetics leveraging smart gloves with IMU and flexion sensors. We demonstrated that the gain from the anticipation of up to several hundred milliseconds could compensate for computing and transmission latency, enabling immersive interactions over extremely long distances. Our solution is flexible and is free from locking into particular smart glove vendors, making it easy for future extensions.

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CCS CONCEPTS

• **Human-centered computing** → **Gestural input**; *Virtual reality*; • **Computing methodologies** → Neural networks.

KEYWORDS

gesture recognition; anticipatory computing; human-machine interactions; deep learning

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1 INTRODUCTION

Immersive technologies such as virtual (VR), augmented (AR), and mixed (MR) reality interfaces are increasingly permeating into leisure and industrial settings as “mainstream” applications. Current setups often rely on head-mounted displays (HMDs), which provide audiovisual interfaces and typically include motion and field-of-view sensing. HMDs are typically paired with hand-based controllers with click buttons that might include additional vibration feedback and feature different sensing capabilities. As pervasive hand-based controllers only allow a minimal and non-intuitive way, e.g., to grasp objects or remotely control robots, smart gloves have been increasingly made commercially available in recent years. Smart gloves are a type of garment that incorporates sensors that capture the motion of the hand and individual fingers and can provide a haptic stimulus to the user using, e.g., vibration motors. Smart gloves allow for more natural and realistic interaction with real and virtual worlds [2].

Providing an immersive audiovisual environment results in enormous demands on communication systems to provide high-fidelity videos with high bitrates and extremely low latency for haptic feedback. While the control and potential feedback information for any hand-based controller are significantly smaller than the audiovisual data, the latency challenge remains to provide immersive virtual or real-world interactions. For example, consider an *immersive* metaverse application where a user interacts with real remote or entirely virtual objects, but a remote server coordinates the overall application. As illustrated in Figure 2, the overall latency experienced between user actions and experienced feedback consists of several components. In turn, a noticeable delay in the interaction, e.g., lifting an object and experiencing feedback, would break the immersive nature of the application. The Tactile Internet [6] focuses on tactile human perception support, which requires around 1 ms service latencies to be fully immersive from a latency viewpoint. However, simple network propagation delays commonly limit the overall range within the 1 ms time boundary. Supporting an immersive, tactile human-machine interface requires new transdisciplinary approaches to solve the inherent latency challenge.

We consider an anticipatory computing approach to solve the challenge, exploiting that psychology and neuroscience have shown that human intention can be anticipated, primarily by evaluating

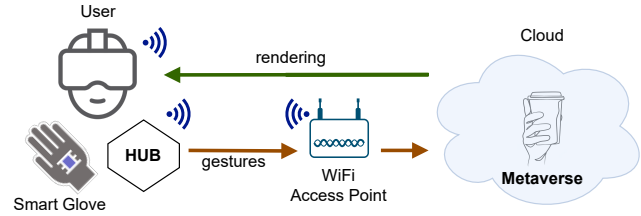


Figure 2: Typical interactions in metaverse between a user and virtual object via AR/VR goggles and smart gloves, including a body computing hub (HUB): Latency incurs in both uplink (i.e., sending recognized hand gestures to the metaverse) and downlink (i.e., rendering updated scenes).

hand kinesthetics. We therefore envision the following use case of our introduced anticipatory system: A system observes the sensor data of wearables, worn by a user that is acting in the metaverse. Based on anticipated intentions of the user, communication networks can prepare proactively to the requirements demanded by the user intention, such as ultra-low latency for a tactile stimulus, reliability or security for a human to human interaction. Our contribution is two-fold. We present a demonstration to illustrate the proof-of-concept and an evaluation of anticipatory computing. The experiments exploit the ex-ante data observed from a smart glove human-machine interface to anticipate human action using user-dependent trained grasping action models. We showcase that the anticipation significantly reduces latency to enable perceived real-time interaction over long distances.

2 DEMO SETUP AND EXPERIMENTATION

2.1 Hardware Setup and Data Visualization

To capture hand gestures, we employ the Manus Meta Prime 2 glove, which was designed for AR and VR application scenarios and utilizes a combination of inertial measurement unit (IMU) and flex-resistive sensors. Five IMUs are positioned at the upper finger joints and the sixth IMU at the back of the hand. Resistive sensors are placed inside each finger sleeve to measure the flexion of the fingers. We sample the orientation data and the flexion of the finger joints with a sampling rate of 50 Hz. We utilize a force resistive sensor (FSR) with a surface area of 116 mm x 76 mm to sense and timestamp the exact moment an object was touched. The resistance of the FSR is read via the analog pin of an Arduino Uno at approximately 100 Hz and transmitted to the laptop via the serial interface (baud rate of 9600 bps). We used a laptop with an AMD Ryzen 7 PRO 4750U processor for data collection and gesture recognition. We chose to incorporate eight objects commonly found in daily activities that are meaningful while interacting in the metaverse (e.g., kitchen environment). The objects are depicted in Fig. 3. Real-time plotting leverages PyQtGraph version 0.13.4 [18] to render the visual overview illustrated in Fig. 1. The visualization consists of four main sections. The upper section of the screen shows a bar plot representing the confidence scores of the anticipation model. The second window displays the anticipated object once the model exceeds a confidence threshold of 0.5. A third screen serves as the qualitative visualization of the data fed into the model, and a fourth

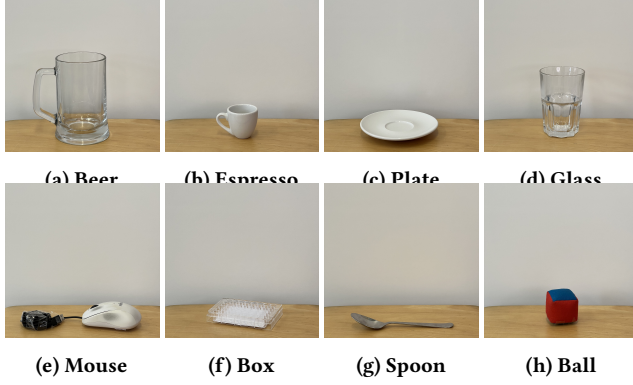


Figure 3: Objects employed in the demo and experimental setup.

plot depicts the time of anticipation ahead of the event of lifting the anticipated object.

2.2 Technology Under the Hood: Anticipation Model

Anticipating grasping events differs inherently from the commonly investigated “static” classification tasks where a data point to be classified is assigned a class label. Anticipation of constantly changing hand postures is dynamic, as the hand evolves from an undefined hand posture (e.g., loosely hanging to the side) to a clearly defined grasp (e.g., holding a ball). Thus, anticipation consists of the interplay between a prediction and the corresponding entailed uncertainty of that prediction. Therefore, we design the model to satisfy two goals. First, the model should accurately anticipate the grasp. Second, the model should also communicate its uncertainty in the case of no grasping motion being performed. Thus, the model exhibits uncertainty in its prediction as the hand is in an undefined posture, not indicating any future grasping event. However, the model becomes progressively more confident as the hand reaches toward a defined object resulting in a grasp. To achieve the desired behaviour of our anticipation model, we first treat the prediction of a grasp as a static classification problem by labelling the entire reach-grasp motion as the end grasp. Then, we calibrate the model to produce uncertain predictions when the reach-grasp motion does not excite indicative hand patterns but produces highly confident predictions as the reach-grasp motion progresses and excites more distinguishable patterns.

To anticipate grasps, we exploit neural networks and implemented a lightweight MobileNet [7] with approximately 20k trainable parameters using the TensorFlow v2.12.0 framework [5]. At each time step, the orientation represented as quaternions of four finger joints relative to the back of the hand is sampled from the glove, which results in a total of 20 hand joint orientations. The previous 10 time steps form the data windows used to train the model. We train the model on a batch size of 2 using the Adam optimizer [10]. We perform data augmentation to increase the ability to generalize on unseen subject data by artificially rotating the obtained orientation estimates and slicing the sensor data randomly into windows of 10 consecutive time steps. Moreover, we calibrate

the model to obtain reliable estimates of the model uncertainty by ensemble-averaging 10 individually trained models [11]. Deriving an ensemble of multiple diverse models is achieved by using probabilistic weight initialisation as well as the inherent randomness of sampling random batches during training via stochastic gradient descent.

2.3 Experimental Data Collection

The set of objects was selected with the aim of loosely resembling the acquisition protocol from the Ninapro Project [9] which aims to incorporate hand movements encountered in daily living. We initially asked two subjects to arbitrarily choose an object from those depicted in Fig. 3 and subsequently grasp, lift, and place the object away from its original spot. We instructed subjects to grasp the object as if they intended to use it for the corresponding purpose (i.e., lift the glass with one as if they intended to drink). For each object, the grasp, lift, and place procedure is repeated for one minute, followed by a calibration of the smart glove. Subjects are encouraged to progress at their own speed. For the two subjects we used in our study, this resulted in approximately 10-12 reach-to-grasp motion recordings for each object. Following the *eigengrasp* notion described in [19] and discussed in Sec. 4, we anticipate that the gathered data for different objects is very similar at the beginning of the reach-to-grasp motion, but excites different hand motion patterns as the reach progresses.

3 EVALUATION RESULTS

This section describes our initial results obtained for subject-specific trained model deployment. Aside from various sources of latency occurring during computation and networking, we highlight two types of latencies. One latency is the inference latency which is the computing time of the anticipation model to derive an anticipation decision. Moreover, we define the “negative latency” t_{nl} as a gain of time ahead of an event that can be used to compensate for latency arising during other networking and computation stages. In the case of our demo, we, therefore, define the “negative latency” t_{nl} as the difference $t_{nl} = t_{grasp} - t_{ant}$ between the moment of object interaction t_{grasp} and the moment of the anticipation model correctly predicting the grasped object t_{ant} .

We initially consider the inference latency required for the trained model to provide its result. We sample data from the smart glove at 50Hz, i.e., one data sample every 20 ms. If the inference latency is too high, there would be only a limited possibility of attaining an overall negative latency through computed anticipation. We execute the anticipation model within a single thread. To evaluate the accuracy of our anticipation model, we run a total of 10 grasping trials for each person and object to derive the 0 ms negative latency classification accuracy. While for most objects the accuracy is above 90%, the objects spoon and mouse can only be recognized with an accuracy of 50-80%. We initially observe that the measured inference latency exceeds 0.5 ms in only occasional cases but never surpasses 1 ms. Therefore, the inference time justifies a real-time model deployment if the attainable negative latency through the computed anticipation can offset the inference latency and is provided at a rate higher than the smart glove sampling frequency.

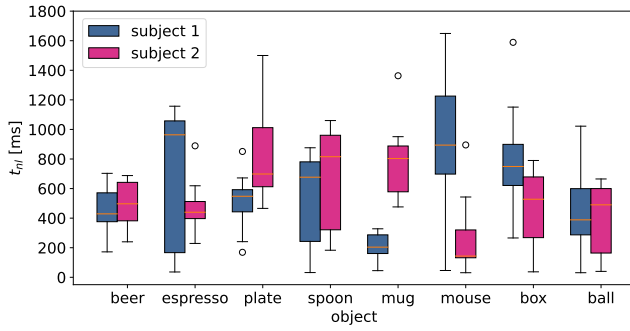


Figure 4: Distribution negative latency for the model fitted to subject-specific data.

We initially collected data for each subject individually and fitted the data to a subject-specific anticipation model. Subsequently, we experimented with grasping different objects, measuring t_{nl} . Differently put, we performed the training/validation before we employed the model in a second round of tests. We timestamped the moment t_{ant} by detecting when the model confidence exceeds a threshold of 0.5. We also timestamped the moment of grasp t_{grasp} by detecting a weight change of the object pressing on the force-resistive sensor. For readability, we focus on data from two individual subjects and illustrate the distribution of measured t_{nl} in Fig. 4. We immediately observe varying anticipation capabilities of the models. For example, while t_{nl} for the glass are similar between subjects, they widely differ for other objects (e.g., mug or mouse). Moreover, the current implementation’s negative latency exhibits different consistencies for objects and subjects. While the distribution of negative latencies for subject 1 grabbing the beer glass is centred narrowly around 200 ms, the distribution for subject two is shifted and spread out around values of 800 ms.

4 RELATED WORKS

This section summarizes related studies on anticipatory computing and especially anticipating human intention from grasp.

4.1 Anticipatory Computing

Anticipatory computing aims to exploit the intricacies of systems, especially where one can consider humans in the loop, to derive predictions of future events [15, 16]. This interdisciplinary research field includes human behaviour analysis, human activity recognition, research in computer architectures, processing storage, and data analytics [17]. The goals of anticipatory computing on the intersection of ubiquitous computing, sensing/actuating, and psychology are manifold [4] and range from user-adaptive information displays [3] to collaborative robotics [8]. Current approaches typically include some form of forecasting of human intention from (wearable) sensors [21] at the human-machine interface. A more recent survey [13] focuses on latency compensating techniques in networks for gaming, describing different methods to reduce latency, including anticipation of players’ actions and inputs. In this contribution, we focus on analysing human motion trajectories and grasping motions using smart gloves as one example, as this type of human-machine interface can intuitively be used in many

different application scenarios. We note that our approach will also apply to other human-machine interfaces.

4.2 Intention Anticipation from Grasps

Psychology research and Neuroscience have studied human hands’ kinesthetic motion because humans rely extensively on their hands to interact with their environments. Studies [14] and [19] investigated motor muscle synergies during grasping motions from grasping movements for various objects of daily living. Their findings show that hand kinematics consist of a few “eigengrasps”. Combinations and augmentations from a common subset of *eigengrasps* resemble the most common grasps of individual objects in daily scenarios. Vice versa, the object to be grasped can be determined based on the *eigengrasps* and the object-specific augmentations.

Additional refinements to this approach include reducing required data points and incorporating additional hints in the object determination process. For example, most hand gesture information holds inside the kinematic information of the thumb and index finger, which could yield less invasive and bulky wearables to sense hand kinematics [12]. Past works from psychology have focused on whether it is feasible to infer the social intent of grasping an object from observing patterns of the hand’s kinematics. While the hand’s kinematics excite different patterns when the same object is grasped with a different intention [1], observing the hand and arm motion is sufficient to infer the performed action [20]. In [22], the authors demonstrate that different hand gestures used in sign language can be efficiently recognised before someone finishes the gesture. Our approach does not include any additional hints and solely relies on the data gathered by the smart glove itself.

5 CONCLUSION & FUTURE WORK

We aim to use anticipatory computing for AR/VR goggles and smart gloves. We presented a novel demonstrator showcasing anticipatory machine learning models to produce predictions of future events based on real-time sensor data. Additionally, we developed a testbed to assess the model’s accuracy and anticipation gain. As our preliminary findings show, different grasps are subject-specific. Thus, user-specific anticipation models are required. In future work, we plan to integrate various sensor modalities and combinations aside from inertial and flex sensors, such as electromyography from the forearm muscles, since our setup’s wearable hardware is interchangeable. Moreover, we plan to collect a large scale dataset of subjects and objects, allowing to test and compare different anticipation models. Moreover, we want to investigate to which extent different but similar grasps can be discriminated.

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REFERENCES

- [1] Caterina Ansuini, Livia Giosa, Luca Turella, Gianmarco Altoè, and Umberto Castiello. 2008. An object for an action, the same object for other actions: effects on hand shaping. *Experimental Brain Research* 185 (2008), 111–119.
- [2] Manuel Caeiro-Rodríguez, Iván Otero-González, Fernando A Mikic-Fonte, and Martín Llamas-Nistal. 2021. A systematic review of commercial smart gloves: Current status and applications. *Sensors* 21, 8 (2021), 2667.
- [3] Cristina Conati, Giuseppe Carenini, Dereck Toker, and Sébastien Lallé. 2015. Towards user-adaptive information visualization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 29.
- [4] Alia El Bolock, Jailan Salah, Yomna Abdelrahman, Cornelia Herbert, and Slim Abdennadher. 2018. Character computing: computer science meets psychology. In *Proceedings of the 17th International Conference on Mobile and Ubiquitous Multimedia*. 557–562.
- [5] Martin Abadi et al. 2015. TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems. <https://www.tensorflow.org/>. Software available from tensorflow.org.
- [6] Frank H. P. Fitzek, Shu-Chen Li, Stefanie Speidel, Thorsten Strufe, Meryem Simsek, and Martin Reisslein (Eds.). 2021. *Tactile Internet with Human-in-the-Loop*. Academic Press. later on under: <https://ceti.one/book>.
- [7] Andrew G Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. 2017. Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861* (2017).
- [8] Chien-Ming Huang and Bilge Mutlu. 2016. Anticipatory robot control for efficient human-robot collaboration. In *2016 11th ACM/IEEE international conference on human-robot interaction (HRI)*. IEEE, 83–90.
- [9] Néstor J Jarque-Bou, Manfredo Atzori, and Henning Müller. 2020. A large calibrated database of hand movements and grasps kinematics. *Scientific data* 7, 1 (2020), 1–10. <https://doi.org/10.1038/s41597-019-0349-2>
- [10] Diederik P Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [11] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. 2017. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in neural information processing systems* 30 (2017).
- [12] Chen Liang, Chun Yu, Yue Qin, Yuntao Wang, and Yuanchun Shi. 2021. DualRing: Enabling subtle and expressive hand interaction with dual IMU rings. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 3 (2021), 1–27.
- [13] Shengmei Liu, Xiaokun Xu, and Mark Claypool. 2022. A survey and taxonomy of latency compensation techniques for network computer games. *ACM Computing Surveys (CSUR)* 54, 11s (2022), 1–34.
- [14] Carolyn R Mason, Jose E Gomez, and Timothy J Ebner. 2001. Hand synergies during reach-to-grasp. *Journal of neurophysiology* 86, 6 (2001), 2896–2910.
- [15] Mihai Nadin. 2016. *Predictive and Anticipatory Computing*. CRC Press. <https://doi.org/10.1081/E-ECST2-120054027>
- [16] Mihai Nadin. 2018. Anticipatory Computing. *Ubiquity* 2000, December (sep 2018), 2–es. <https://doi.org/10.1145/357518.357520>
- [17] Veljko Pejovic and Mirco Musolesi. 2015. Anticipatory mobile computing: A survey of the state of the art and research challenges. *ACM Computing Surveys (CSUR)* 47, 3 (2015), 1–29.
- [18] Riverbank Computing. [n. d.]. *PyQtGraph*. <https://www.pyqtgraph.org/>
- [19] Marco Santello, Martha Flanders, and John F Soechting. 1998. Postural hand synergies for tool use. *Journal of neuroscience* 18, 23 (1998), 10105–10115.
- [20] Luisa Sartori, Cristina Becchio, and Umberto Castiello. 2011. Cues to intention: the role of movement information. *Cognition* 119, 2 (2011), 242–252.
- [21] Julian Steil, Philipp Müller, Yusuke Sugano, and Andreas Bulling. 2018. Forecasting user attention during everyday mobile interactions using device-integrated and wearable sensors. In *Proceedings of the 20th international conference on human-computer interaction with mobile devices and services*. 1–13.
- [22] Peter Vamplew and Anthony Adams. 1995. Recognition and anticipation of hand motions using a recurrent neural network. (1995).